

MamaCare AI - Slide-by-Slide Deep Dive

Slide 2 - Problem and Context

What the slide says

Headline quote:

"800 women die every day from preventable causes related to pregnancy and childbirth." - WHO
Maternal Mortality Fact Sheet 2025

Core problem: "The Care Gap"

- Maternal deaths aren't a single event - they are **missed signals across the entire pregnancy journey**
- Only **1 in 3 women** in rural Sub-Saharan Africa has mobile internet (GSMA 2023)
- Most women have only a **basic phone** (voice calls + SMS only)

"Meet Abena"

- A typical pregnant woman in this context gets **little to no structured support between clinic visits**
- This leads to **unseen physical and mental health crises** in between appointments

Key terms

Term	What it means
WHO	World Health Organization - the UN agency that sets global health standards
Maternal Mortality	Death of a woman while pregnant or within ~42 days of giving birth, from causes related to the pregnancy
800 women per day	~287,000 per year globally - the statistic from WHO
Sub-Saharan Africa	The part of Africa south of the Sahara Desert (where most of our target users are)

Term	What it means
GSMA	Global System for Mobile Communications Association - the industry body for mobile networks worldwide
Mobile internet	Smartphone-style internet access (needs data plan, smartphone, signal). Different from basic GSM voice which only needs a simple phone
Prenatal / Antenatal	The period before birth - during pregnancy
Basic phone / Feature phone	A simple phone that can make calls and send texts but cannot run apps or browse the web
Continuous structured support	Regular, organized check-ins and monitoring - not just when the woman happens to visit a clinic

What this slide is doing

It builds the case for why our product exists. The argument flows like this:

1. **Big scary number** (800 women/day) → attention grabber
2. **Why this happens** (missed signals between visits) → the real problem
3. **Why existing digital health fails** (requires internet/smartphone that most women don't have) → the gap
4. **Abena** puts a human face on the statistic → emotional connection

The slide ends on a cliffhanger - "unseen crises." The next slide answers: *here's what we built to fix that.*

Slide 3 - Solution

What the slide says

Headline:

A Voice-based Conversational AI Agent for prenatal risk monitoring and support for perinatal loss mothers (Zero-Friction Innovation)

Two user groups with features:

For Pregnant Women	For Perinatal Loss Mothers
Automated check-in calls in her language that triage symptoms	Counselor support services and AI-powered mental wellness resources
Support for local language	Scheduled check-ins for the bereaved mother
No app, no data - 24/7 accessibility	

Key terms

Term	What it means
Voice-based Conversational AI Agent	A computer system that talks to you over a phone call like a human would. You speak, it listens, it responds. Think of an automated assistant that uses voice, not text
Prenatal risk monitoring	Watching for signs of danger during pregnancy (bleeding, high blood pressure, fever, etc.)
Perinatal loss	The death of a baby around the time of birth - includes miscarriage (pregnancy loss before 20 weeks), stillbirth (baby dies in womb after 20 weeks), or neonatal death (baby dies shortly after birth)
Zero-Friction Innovation	A solution that removes all barriers to use. Here: no app to download, no internet needed, no smartphone required, no literacy requirement. Just pick up the phone
Triage symptoms	Sort the woman's reported symptoms by urgency - LOW (routine advice), MEDIUM (needs review within 24h), HIGH (immediate action needed)
AI-powered mental wellness resources	The AI suggests coping strategies, support messages, or connects to counselor support for mothers who have experienced loss
Scheduled check-ins	The system automatically calls the mother at planned times - she doesn't need to remember to reach out
24/7 accessibility	Available any time of day or night, any day of the week

What this slide is doing

This slide answers the problem. After Slide 2 showed women have only a basic phone and no support between visits, this slide says:

- **Solution = voice calls** (works on basic phones)
- **Solution = in her language** (no literacy barrier)
- **Solution = automated** (no extra work for health workers)
- **Solution = two use cases** (pregnancy + post-loss - covers the full journey)

The phrase "Zero-Friction Innovation" signals to judges that we designed for the **hardest case** (poorest, least connected woman) - not for the easiest.

Slide 4 - Solution: Clinical Dashboard (Core SaaS Component)

What the slide says

Module	Features
Risk Overview	Live health risk escalation feed
User Management	Activity log
Personalised & Mass Health Alerts	Medication reminders, nutrition & mental wellness tips

Key terms

Term	What it means
SaaS (Software as a Service)	Software that runs on the cloud and is accessed through a web browser. Users don't install it - they just log in. Like Gmail or Google Docs
Clinical Dashboard	A screen that health workers look at to see all their patients' status at a glance
Risk escalation feed	A live, updating list of patients sorted by urgency. A new HIGH-risk case appears at the top immediately

Term	What it means
Activity log	A record of every action - when a patient was called, what was reported, what action was taken. Like a digital diary
Personalised Health Alerts	Messages sent to individual patients based on their specific situation (e.g., "take your iron supplement")
Mass Health Alerts	Messages sent to all patients at once (e.g., "malaria outbreak in your district - sleep under a mosquito net")
WebSocket alerts	Real-time notifications - when a HIGH-risk case comes in, the dashboard updates instantly without the page needing to refresh. Like a live sports score update

What this slide is doing

This slide introduces the **other half** of the product. Slide 3 was about the patient experience (voice calls). This slide is about the **health worker experience** (dashboard). Together they make a complete system:

- Voice agent catches symptoms → dashboard alerts the clinician → clinician acts

The three modules cover the full workflow:

1. **Risk Overview** - what needs attention now
2. **User Management** - who are my patients, what's their history
3. **Health Alerts** - how do I communicate with them

Slide 5 - Methodology, Ethical and Responsible AI

What the slide says

Pipeline (4 steps):

1. **National Safe Motherhood Protocol** (Ghana Health Service) - the source document
2. **Clinical Triage Guidelines** - 28 initial samples, oversampled to 400
3. **Guideline-Derived Few-Shot Fine-Tuning** - distilbert-base-multilingual-cased (HuggingFace)

4. Testing and Evaluation

Model Performance:

Overall Test Accuracy: **66.67%**

Risk Level	Precision	Recall	F1
HIGH	0.50	1.00	0.71
MEDIUM	1.00	0.20	0.33
LOW	0.80	0.80	0.80

Key terms

Term	What it means
National Safe Motherhood Protocol	The official Ghana Health Service document that defines what counts as a danger sign in pregnancy and what to do about it. Think of it as the rulebook for maternal care in Ghana
Ghana Health Service	The government agency that runs public healthcare in Ghana
Clinical Triage Guidelines	The rules we extracted from the Safe Motherhood Protocol that tell us: "If symptom X, then risk level Y"
28 initial samples	We started with 28 example symptom descriptions (e.g., "heavy bleeding after birth" → HIGH risk) as our seed data
Oversampled to 400	We duplicated and varied those 28 examples to create 400 training samples. This is common when you have very little real patient data - you make more data from what you have
Guideline-Derived Few-Shot Fine-Tuning (GDF-FT)	A fancy way of saying: we took an existing AI model and taught it to recognize pregnancy danger signs by feeding it the Safe Motherhood Protocol rules. "Few-shot" means we taught it with very few examples
distilbert-base-multilingual-cased	A compact AI language model from HuggingFace that understands 100+ languages. "Distil" = distilled (smaller, faster version). "Multilingual" = works in many languages. "Cased" = cares about capital letters ("Apple" vs "apple")

Term	What it means
HuggingFace	A platform where AI researchers share pre-trained models - like GitHub for AI
Precision	Of all the cases the model labeled as HIGH, what fraction were actually HIGH? Precision = how often the model is <i>correct when it flags something</i>
Recall	Of all the actual HIGH-risk cases out there, how many did the model catch? Recall = how <i>thorough</i> the model is
F1	A combined score that balances Precision and Recall. 1.0 = perfect, 0.0 = terrible
Support	The number of test samples in each class (we haven't displayed it here, but in the Technical Document it was 5 per class)

How to read the performance table

Class	Precision	Recall	F1	What it tells us
HIGH	0.50	1.00	0.71	Caught every HIGH case (Recall = 1.00) but flagged some extra non-HIGH cases as HIGH (Precision = 0.50 - half of its HIGH flags were wrong)
MEDIUM	1.00	0.20	0.33	When it said MEDIUM it was always right (Precision = 1.00) but it missed 80% of actual MEDIUM cases (Recall = 0.20)
LOW	0.80	0.80	0.80	Most balanced - caught 80% of LOW cases and was correct 80% of the time when it said LOW

Why this slide is important

The 66.67% accuracy is not great - and the Technical Document itself says "we acknowledge that more work needs to be done." But we present it transparently because:

1. We started with only **28 examples** - no other team built a triage model from scratch with that little data
2. **HIGH Recall = 1.00** - the model catches every single HIGH-risk case. It never misses a danger. Some false alarms are acceptable in a triage setting (better to over-call than under-call)

3. The methodology (GDF-FT) is innovative - encoding clinical guidelines into a model is a real research contribution

Slide 6 - Data Use, Open Science and Reproducibility

What the slide says

Four bullet points:

- **National Safe Motherhood Protocol - MoH** (Openly available)
- **DistilBERT Model** (Open source)
- **Open Source Stack** for Clinical Dashboard (React, Express.js, PostgreSQL)
- **Code available on GitHub** with MIT Licence

And a summary sentence:

The project complies with Open Data for Research, use of open source models and stack for Dashboard Web App development, and availability of reproducible code.

Key terms

Term	What it means
MoH	Ministry of Health (Ghana's government health department)
Openly available	Anyone can access it - no paywall, no special permission needed
Open source	The code is publicly viewable, usable, and modifiable by anyone. Like a recipe anyone can see and adapt
React	A popular JavaScript library for building user interfaces (the frontend)
Express.js	A web framework for Node.js that handles server-side logic (receiving requests, sending responses)
PostgreSQL	A powerful open-source database - stores patient records, call logs, risk assessments

Term	What it means
MIT Licence	A very permissive open-source license. Anyone can use, copy, modify, and distribute our code, even commercially, as long as they include the original copyright notice
Reproducible code	Code that someone else can run on their computer and get the same results. Essential for scientific credibility

What this slide is doing

This slide shows judges we are **good open science citizens**. In research competitions (especially HASH), openness and reproducibility are highly valued because:

1. **Others can verify our results** - they can run the same code
2. **Others can build on our work** - the MIT license means researchers or health systems can adapt MamaCare for their own context
3. **We used public data** - no proprietary datasets that can't be shared
4. **The whole stack is open** - no vendor lock-in, no expensive licenses needed

Slide 7 - Target and Value Proposition

What the slide says

Three user segments and what each gets:

Segment	Value Delivered
Pregnant & Prenatal Loss Mothers	Regular voice check-ins (no app needed), continuous symptoms monitoring
Health Facilities	AI triages routine cases, effective digital records
District Health Admin & Government	Real-time KPIs for analytics, exportable reports for planning

Key terms

Term	What it means
Prenatal Loss	Loss of a baby before birth (miscarriage, stillbirth)
Continuous symptoms monitoring	The system tracks symptoms over time - not just one-off checks. It can see if a headache is getting worse day by day
AI triages routine cases	The AI handles the easy/normal cases automatically, freeing health workers to focus on the serious ones
Effective digital records	Instead of paper registers that get lost or damaged, patient data is stored digitally and is searchable
KPIs (Key Performance Indicators)	Numbers that tell administrators how the system is performing - e.g., number of high-risk cases detected, average response time, number of mothers enrolled
Exportable reports	Administrators can download data as reports (e.g., spreadsheets) for planning and budgeting

What this slide is doing

This slide shows that we understand **all the stakeholders** - not just the mother, but the people who would pay for and operate the system:

- **Mothers** get the service for free, no barriers
- **Health Facilities** get efficiency (AI handles routine triage) and better record-keeping
- **Government/Admin** gets data for planning - they can see which districts need more resources

The message to judges: "We know who needs what, and we deliver value to everyone in the chain."

Slide 8 - Impact

What the slide says

SDG Alignment:

- SDG 3 - Good Health and Well-Being
- SDG 5 - Gender Equality
- SDG 10 - Reduced Inequalities

Three Key Impact Areas:

Impact	What it means
Reduced Maternal Mortality	Voice agent check-ins catch danger signs between clinic visits before they become fatal
Increased Maternal Health Service Productivity	The system enables health institutions to manage large numbers of prenatal mothers at increased efficiency
Bridges Infrastructure Barrier	Over 90% GSM reachable - continuous maternal care accessible to anyone regardless of geographical location and resources

Key terms

Term	What it means
SDG (Sustainable Development Goals)	17 global goals set by the United Nations in 2015 to achieve a better and more sustainable future by 2030
SDG 3: Good Health and Well-Being	Ensure healthy lives and promote well-being for all. Target 3.1 specifically: reduce global maternal mortality to less than 70 per 100,000 live births
SDG 5: Gender Equality	Achieve gender equality and empower all women and girls. Maternal health is a gender equality issue - women die because systems don't serve them
SDG 10: Reduced Inequalities	Reduce inequality within and among countries. Poor rural women die at much higher rates than wealthy urban women. MamaCare targets the most underserved
GSM (Global System for Mobile Communications)	The standard cellular network technology that basic phones use. "90% GSM reachable" = 9 out of 10 people can get a signal for voice calls

Term	What it means
Maternal Mortality	Death caused by or aggravated by pregnancy or its management
Health Service Productivity	How much care a health system can deliver with its available resources (staff, facilities, budget)

What this slide is doing

This slide connects our project to **global frameworks that funders and governments care about**. By showing SDG alignment, we tell judges:

- Our project is not just a clever tech demo - it directly contributes to **measurable global goals**
- Funding maternal health is not charity - it's **SDG commitment**
- We can measure our impact against these targets

The three impact areas cover the main arguments:

1. **Reduced deaths** - the ultimate goal
2. **Better use of health workers** - efficiency argument
3. **Works where nothing else does** - the 90% GSM reachability argument

Slide 9 - Next Steps: Scalability and Sustainability

What the slide says

Three next steps:

1. **Language expansion** - implement support for Twi, Fante, Ga, and Dagbani
2. **Model improvement** - extensive guideline anchors and new training strategies
3. **System integration** - integration into the Ghana Health System

Key terms

Term	What it means
Twi	A major Ghanaian language spoken by the Ashanti people and widely used across Ghana
Fante	A Ghanaian language spoken by the Fante people in the coastal regions
Ga	A Ghanaian language spoken in and around Accra, the capital
Dagbani	A Ghanaian language spoken in the Northern Region of Ghana
Guideline anchors	More clinical symptom descriptions from the Safe Motherhood Protocol to train the model on - more examples means better accuracy
New training strategies	Different approaches to improving the model - more data, different fine-tuning methods, etc.
Ghana Health System	The full network of public healthcare in Ghana - from community clinics (CHPS compounds) to district hospitals to the Ministry of Health

What this slide is doing

This slide shows judges we have a **plan beyond the prototype**.

1. **We'll cover more languages** - starting with Ghana's four main local languages
2. **We'll improve the model** - we know it needs work, and we have a clear next step
3. **We'll embed in the existing system** - not a standalone app, but integrated into what already exists